

Companion LXXIX: Score-Matching and Diffusion Models for Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

Score-based diffusion models provide a powerful generative framework for learning complex probability distributions through stochastic differential equations. Applied to persistence diagrams, these models learn the gradient of the log-density (score function), enabling sampling, denoising, and likelihood-free inference in topological data analysis. Building on the Schrödinger bridge dynamics introduced in Companion LXXVIII, this work develops score-matching and diffusion models for persistence diagrams in the Relaxation-Driven Cyclic Cosmology (RDCC). The relaxation mechanism modifies the scale-dependent evolution of the gravitational potential, inducing characteristic changes in score fields, diffusion trajectories, and generative flows. The resulting framework provides a modern, flexible, and probabilistically grounded approach to modeling RDCC signatures in weak-lensing topology.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. Previous Companions introduced Wasserstein geometry, entropic optimal transport, and Schrödinger bridges as tools for analyzing the Relaxation-Driven Cyclic Cosmology (RDCC).

Score-based diffusion models extend this framework by learning the gradient of the log-density of persistence diagrams. This score function governs the reverse-time diffusion process that generates new diagrams from noise. In cosmology, this provides a powerful method for modeling the distribution of topological features, denoising weak-lensing maps, and performing likelihood-free inference.

In RDCC, the relaxation mechanism modifies the underlying distribution of persistence diagrams, producing characteristic signatures in the score field and the associated diffusion dynamics.

2 Score Matching for Persistence Diagrams

Let $p(D)$ be the probability density of persistence diagrams. The score function is

$$s(D) = \nabla_D \log p(D). \quad (1)$$

Score matching estimates $s(D)$ by minimizing

$$\mathcal{L}_{\text{SM}} = \mathbb{E}_p \left[\|s_\theta(D)\|^2 + 2 \nabla \cdot s_\theta(D) \right], \quad (2)$$

where s_θ is a neural approximation of the score.

For persistence diagrams, the gradient acts on birth-death coordinates.

3 Diffusion Models for Persistence Diagrams

Forward diffusion adds noise:

$$dD_t = \sqrt{\beta(t)} dB_t, \quad (3)$$

where B_t is Brownian motion on diagram space.

The reverse-time SDE is

$$dD_t = [-\beta(t)s(D_t, t)] dt + \sqrt{\beta(t)} d\bar{B}_t, \quad (4)$$

where $s(D_t, t)$ is the time-dependent score.

Sampling proceeds by integrating this reverse SDE from noise to data.

4 RDCC Modification of Score Fields

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_\kappa \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (5)$$

This modifies the log-density of persistence diagrams:

$$\log p_{\text{RDCC}}(D) = \log p(D) - \chi_p \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha. \quad (6)$$

The score becomes

$$s_{\text{RDCC}}(D) = s(D) \left[1 - \chi_s \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (7)$$

5 RDCC Diffusion Dynamics

The RDCC reverse-time SDE is

$$dD_t = -\beta(t)s(D_t) \left[1 - \chi_s \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right] dt + \sqrt{\beta(t)} d\bar{B}_t. \quad (8)$$

RDCC induces:

- reduced score magnitude at small scales,
- enhanced score magnitude at large scales,
- modified diffusion trajectories,
- altered generative flows,
- scale-dependent deformation of reverse-time dynamics.

6 Conclusion

Score-based diffusion models provide a modern and flexible framework for modeling the distribution of persistence diagrams. In the Relaxation-Driven Cyclic Cosmology, the relaxation mechanism modifies the score field and the associated diffusion dynamics in a scale-dependent manner, producing distinctive signatures in weak-lensing topology. The present Companion establishes the foundation for future studies of generative modeling, neural SDEs, and diffusion-based inference in cosmological topology.

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